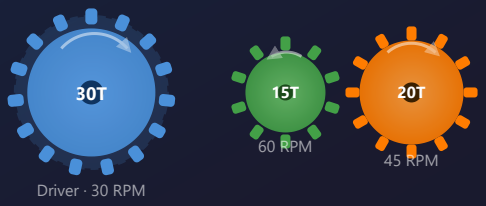
USER MANUAL

3D Gear Simulator

Complete guide to building, connecting, and simulating spur gear trains in 2D and immersive VR.



Spur Gear Physics

Mathematically precise involute tooth profiles with auto-snap meshing

Gear Ratio Engine

Live RPM, ratio, and direction propagation across entire gear trains

VR Mode

Grab and position gears in immersive 3D with Meta Quest hand tracking

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CH 01

Overview

Introduction to the Gear Simulator, its layout, and general navigation.

1.1 What is the Gear Simulator?

The **3D Gear Simulator** is an interactive browser application for building and animating spur gear trains. It uses mathematically precise involute tooth profiles and real gear physics to simulate RPM, gear ratios, and rotation direction through connected gear chains.

Build

Add gears with custom teeth, module, and size. Drag them together to snap and mesh automatically.

Simulate

Set input RPM, press Play. Watch speed and direction propagate through the entire gear train.

Analyse

Read live gear ratios, output RPM, rotation direction, and connection count in the info panel.

1.2 Interface Layout

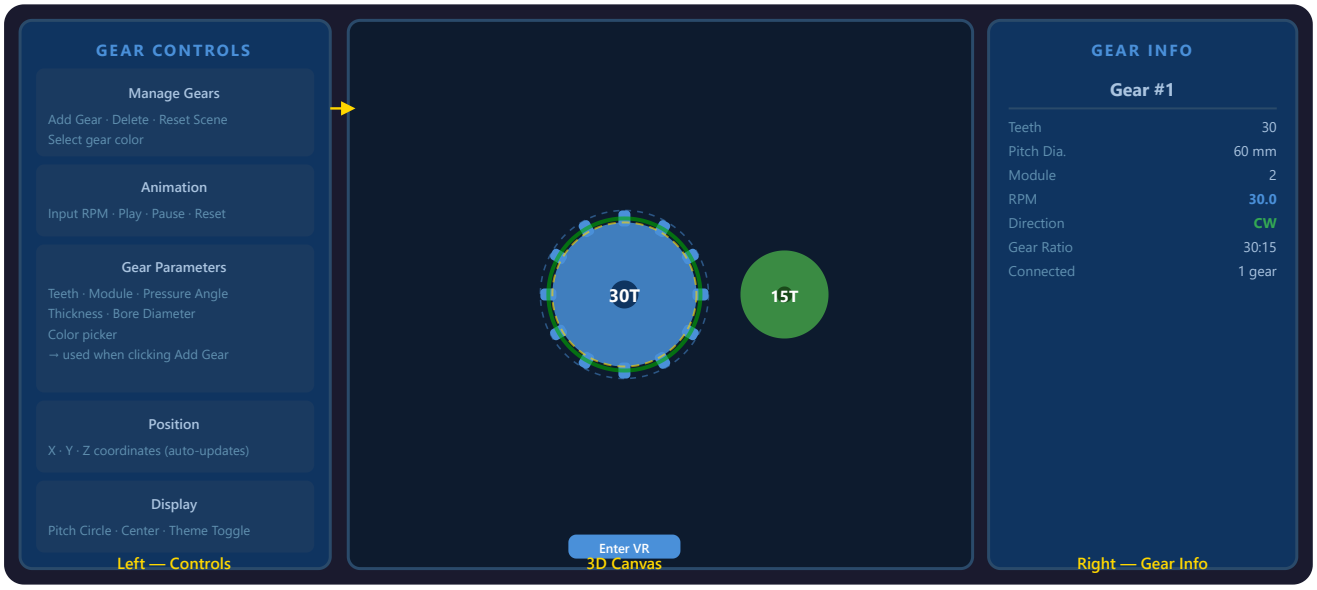


Figure 1.1 — Application layout: left controls panel, 3D canvas with gear train, right info panel.

1.3 Theme Toggle

Click **Toggle Theme** at the bottom of the left panel to switch between **Dark Mode** (default — deep navy background) and **Light Mode** (white/gray background). Both modes preserve all functionality.

CH 02

Gear Fundamentals

Understanding gear parameters, adding gears, and how auto-snap meshing works.

2.1 Gear Parameters

Parameter	Default	Range	What it controls
Teeth	20	8 – 100	Number of teeth. More teeth = larger gear. *Must match module to mesh.*
Module	2 mm	0.5 – 10	Tooth size scale. **Two gears must share the same module to mesh.**
Pressure Angle	20°	14.5 – 25°	Contact angle of teeth. 20° is standard.
Thickness	5 mm	1 – 20	Depth of the gear (extrusion width).
Bore Diameter	5 mm	1 – 20	Diameter of the center hole (shaft size).
Color	#4a90d9	Any color	Visual color of the gear in the 3D view.

Key rule: Two gears can only mesh if their **Module** values are identical (within ±0.001). If modules differ, the snap will be blocked and an error is shown.

2.2 Adding Gears

1. Configure the desired parameters in the **Gear Parameters** section of the left panel.
2. Click **Add Gear**. The new gear appears in the scene, offset from existing gears.
3. The new gear is automatically selected (green emissive glow).
4. Drag it into position. Parameters can be changed before or after adding.

2.3 Connecting Gears (Auto-Snap)

Gears connect automatically when dragged within meshing range. No manual "connect" step is needed.

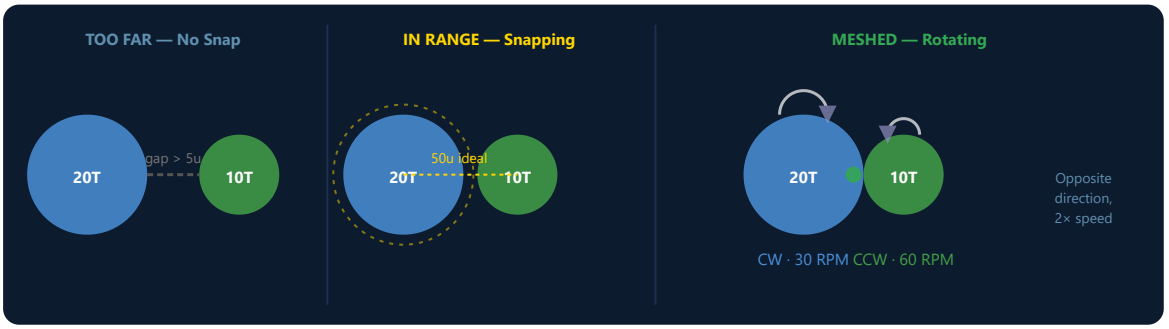


Figure 2.1 — Gear snap sequence: gears too far apart (no snap), in snap range (alignment occurs), fully meshed (opposite rotation, speed ratio applied).

2.4 Gear Ratio & Speed Propagation

When gears are connected, speed propagates through the chain using:

$$RPM_2 = RPM_1 \times (Teeth_1 \div Teeth_2)$$

- **More teeth** on output = slower speed, more torque
- **Fewer teeth** on output = faster speed, less torque
- Meshed gears always rotate in **opposite directions**
- Speed cascades through the entire connected chain (BFS)

Example Chain

Gear	Teeth	RPM	Dir
Driver	30	30	CW
Gear 2	15	60	CCW
Gear 3	20	45	CW

$30 \times (30 \div 15) = 60$ · $60 \times (15 \div 20) = 45$

CH 03

Desktop Controls

Mouse interactions, left panel controls, right info panel, and animation playback.



Figure 3.1 — Left control panel with all sections annotated.

3.1 Selecting Gears

- `Left-click` a gear → selects it (green glow + info panel updates)
- `Left-click` empty canvas → deselects
- Hover over a gear → subtle blue-gray highlight appears

3.2 Moving Gears

- `Left-click + drag` — moves gear in the XY plane
- `Shift + drag` — moves gear along the Z-axis (depth)
- X/Y/Z fields update live as you drag
- Snap check runs on every move; gears auto-connect when close enough

3.3 Camera Controls

- `Middle-mouse drag` — orbit camera around the scene
- `Scroll wheel` — zoom in / out (range: 20 – 500 units)
- Camera damping (0.05) gives a smooth, inertial feel

3.6 Animation Playback

1. Set **Input RPM** (1–1000)
2. Click **▶ Play** — all connected gears start rotating
3. Click **⏸ Pause** — rotation stops, angles are preserved
4. Click **↺ Reset Rotation** — all gears return to 0°

3.4 Display Options

Pitch Circle

Tick **Show Pitch Circle** to draw a dashed green circle around each gear at the theoretical pitch diameter — the ideal meshing contact radius. Turns yellow on the selected gear.

Center Marker

Tick **Show Center** to place a small red sphere at the exact rotational centre of each gear. Useful for verifying gear spacing.

Theme Toggle

Click **Toggle Theme** to switch between dark (navy) and light (white) UI. The 3D canvas adapts its background colour accordingly.

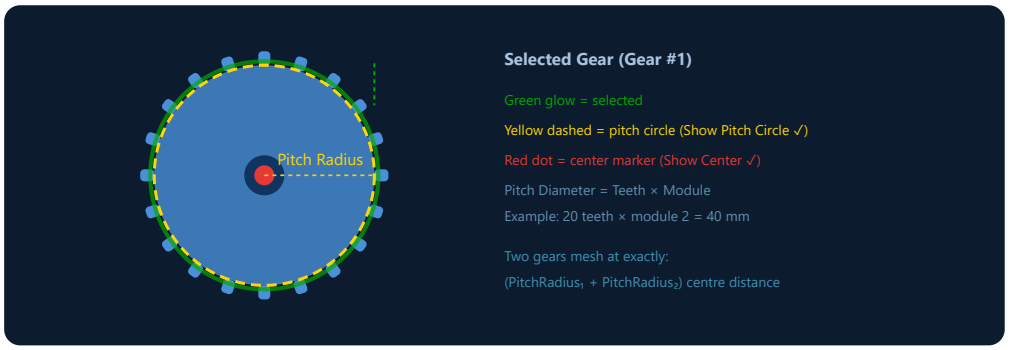


Figure 3.2 — Selected gear with pitch circle (yellow dashed) and center marker (red dot) visible.

3.5 Right Info Panel

Click any gear to see its live data in the **Gear Info** panel on the right:

Field	Description
Gear #ID	Unique identifier assigned when the gear was added
Teeth	Number of teeth as configured
Pitch Diameter	Teeth × Module (mm) — calculated automatically
Module	Current module value
RPM	Live rotation speed (updates during Play)
Direction	**CW** (clockwise) or **CCW** (counter-clockwise)
Gear Ratio	Ratio to each connected gear (e.g. "30:15")
Connected	Number of directly meshed gears

Error states: The info panel background turns **red** (module mismatch), **orange** (jamming), or **purple** (overlap) when a problem is detected.

CH 04

Error States & Feedback

The simulator detects three types of errors and provides visual, textual, and audio feedback for each.

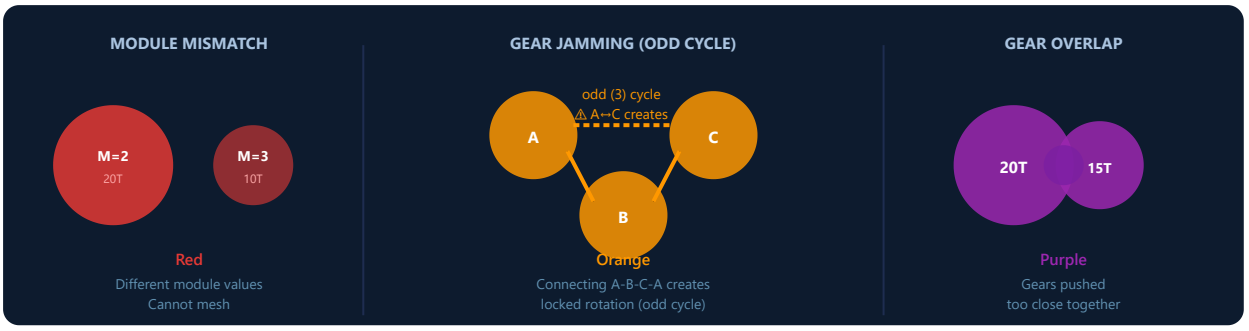


Figure 4.1 — Three error states: module mismatch (red), gear jamming/odd cycle (orange), gear overlap (purple).

4.1 Module Mismatch

- **Cause:** Two gears with different Module values are dragged together
- **Visual:** Gear turns red
- **Message:** "Gears incompatible — modules don't match"
- **Sound:** incompatible-gear.mp3
- **Fix:** Move apart, or change both gears to the same module

4.2 Gear Jamming

- **Cause:** Connecting gears would form a closed loop with an odd number of meshes
- **Visual:** Gear turns orange
- **Message:** "Gear jamming! Locked gear train cycle."
- **Sound:** jammed-gear.mp3
- **Fix:** Move gear to break the cycle. Play is disabled while jamming exists

4.3 Gear Overlap

- **Cause:** Gears pushed closer than 90% of ideal mesh distance
- **Visual:** Gear turns purple
- **Message:** "Gears overlapping! Move gear further apart."
- **Sound:** overlapping-gear.mp3
- **Fix:** Drag gear further away until overlap clears

4.4 Sound Effects

Sound	Trigger
play-gear.mp3	Clicking Play — gear rotation begins
place-gear.mp3	Successful gear snap-to-mesh during drag
incompatible-gear.mp3	Module mismatch detected during drag
jammed-gear.mp3	Jamming / odd cycle detected
overlapping-gear.mp3	Gears pushed too close together
introduction-gear.mp3	Entering VR mode

CH 05

VR / XR Mode

Immersive gear building with Meta Quest hand tracking and 3D floating control panels.

5.1 Entering and Exiting VR

1. Open the app in a WebXR-compatible browser on a Meta Quest headset
2. Click the **Enter VR** button at the bottom-centre of the page
3. The 2D desktop panels hide; 3D floating panels appear in the VR world
4. Intro sound plays; scene brightens for clear visibility in passthrough
5. To exit, use the Quest system menu or the Exit button on the VR panel

VR Requirements

- Meta Quest 2 or Quest 3 (hand tracking required)
- App served over **HTTPS** (WebXR requirement)
- Chrome or Meta Browser on the headset
- Hand tracking enabled in Quest settings

5.2 Hand Gestures

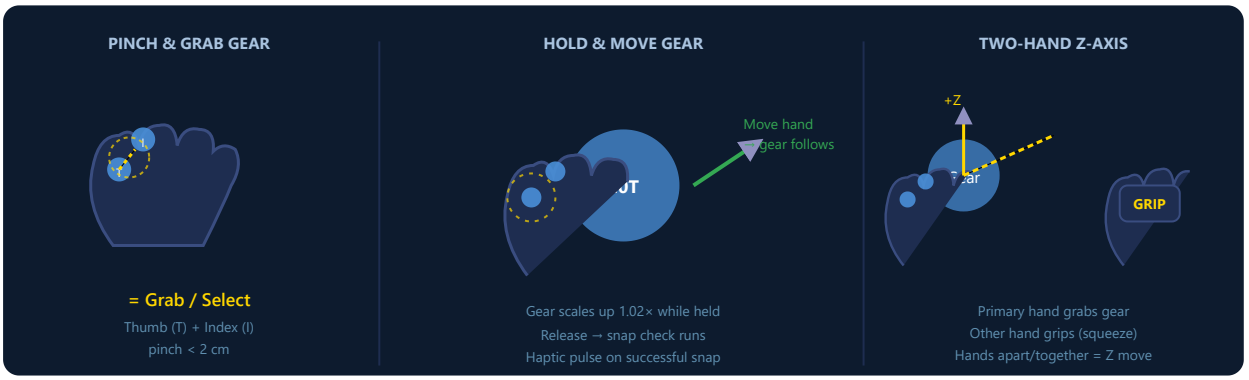


Figure 5.1 — VR hand gestures: pinch to grab, hold and move, two-hand squeeze for Z-axis control.

5.3 VR Floating Panels

Left Control Panel

Positioned left of the gear scene. Contains all the same controls as the desktop — Add Gear, Delete, Reset, Play, Pause, RPM (with ± buttons), and all gear parameters with increment/decrement buttons.

Right Info Panel

Positioned right of the scene. Shows selected gear data live: RPM, direction, ratio, connected count. Background colour changes on error states (red/orange/purple).

Warning Panel

Appears in front of the user when an error is detected. Colour-coded by error type. Auto-hides after ~2 seconds or when the gear is moved away from the problem.

Tip: The **Ray Pointer** changes colour as feedback — blue when pointing at empty space, green when hovering over a gear or panel button. Pinch to activate the hovered target.

CH 06

Quiz Mode

Educational assessment mode — build gear trains to match target specifications within a time limit.

6.1 Activating Quiz Mode

Quiz Mode is activated by adding URL parameters when opening the application:

```
https://[your-server]/?user_id=123&exam_id=456
```

- **user_id** — the student/learner ID used for score submission
- **exam_id** — which exam/question set to load from the server API
- Without these parameters, the app runs in normal free-build mode

6.2 Quiz Flow

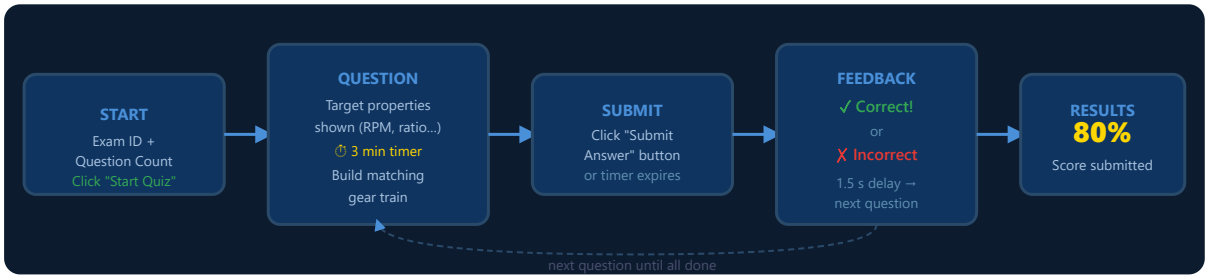


Figure 6.1 — Quiz flow: Start → Question → Submit → Feedback → (repeat) → Final Results.

6.3 Question Types & Validation

Type	Goal	How to solve	Tolerance
gearRatio	Achieve a target gear ratio (e.g. 3:1)	Build chain where driver teeth ÷ output teeth = target ratio	±0.1
rpmOut	Achieve target output RPM	Connect gears so output RPM matches — use RPM × ($T_1 \div T_2$) formula	±0.5 RPM
torqueOut	Achieve target output torque (Nm)	Torque = 10 Nm × (output teeth ÷ driver teeth). Larger output gear = more torque	±0.5 Nm
Legacy (combined)	Match multiple properties at once	Read all required fields (teeth count, module, RPM, ratio) and satisfy each	Per-field

Timer

Each question gives **3 minutes**. The timer colour changes:

- Green — more than 1:30 remaining
- Orange — 30 s to 1:30 remaining
- Red — under 30 seconds

If time runs out the answer is auto-submitted as incorrect.

Results

After all questions, a results screen shows:

- Overall percentage (colour-coded: green >70%, orange 40–70%, red <40%)
- Per-question ✓/✗ breakdown
- Submission status (scores sent to server)

CH 07

Quick Reference

Desktop shortcuts, VR controls summary, and gear parameter reference.

7.1 Desktop Keyboard & Mouse Shortcuts

Action	Input
Select a gear	`Left-click` on gear
Deselect	`Left-click` on empty canvas
Move gear (XY plane)	`Left-click + drag` on gear
Move gear (Z-axis / depth)	`Shift` + `Left-click + drag`
Orbit camera	`Middle-mouse drag`
Zoom camera	`Scroll wheel`
Add gear	Configure params → click **Add Gear**
Delete selected gear	Select gear → click **Delete Selected**
Clear entire scene	Click **Reset Scene**
Start rotation	Set RPM → click ▶ **Play**
Stop rotation	Click ⏸ **Pause**
Reset rotation angles	Click ↺ **Reset Rotation**
Toggle pitch circle	Tick **Show Pitch Circle**
Toggle center marker	Tick **Show Center**
Toggle dark/light theme	Click **Toggle Theme**

7.2 VR Controls Summary

Action	Gesture
Select / activate panel button	Aim ray (green) at button → pinch
Grab and move gear (XY)	Point at gear → pinch + move hand
Move gear along Z-axis	Primary pinch + other hand squeeze → hands apart/together
Release gear (snap check)	Open pinch (fingers apart >4 cm)
Add gear	Aim at **Add Gear** button → pinch
Delete gear	Select gear → aim at **Delete Selected** → pinch
Play simulation	Aim at ▶ **Play** → pinch
Adjust RPM / parameters	Aim at **+** or **−** buttons → pinch

7.3 Gear Parameter Reference

Parameter	Default	Min	Max
Teeth	20	8	100
Module (mm)	2	0.5	10
Pressure Angle (°)	20	14.5	25
Thickness (mm)	5	1	20
Bore Diameter (mm)	5	1	20
Input RPM	30	1	1000

Calculated Values

- Pitch Diameter = Teeth × Module
- Pitch Radius = Pitch Diameter ÷ 2
- Addendum = Module
- Dedendum = 1.25 × Module
- Outer Radius = Pitch Radius + Addendum
- Mesh Distance = $(PR_1 + PR_2)$

Remember: Module must match exactly between two gears for them to mesh. This is the most common reason snap fails.

